

DIGITAL FORENSIC EXAMINATION REPORT

UNMANNED AERIAL VEHICLE (UAV) FLIGHT DATA & TELEMETRY EXTRACTION

PART I: EXECUTIVE INCIDENT SUMMARY (NON-TECHNICAL & JUDICIAL BRIEF)

1. INVESTIGATIVE CASE ADMINISTRATION & PLAIN-LANGUAGE EXECUTIVE SUMMARY

Case Reference:	CASE-GUI-TEST-001	Evidence Item ID:	gui_test_flight.BIN
Examining Agency:	Cyber Forensic Investigation Laboratory (CFSL / State Police)	Lead Examiner:	Test Examiner
Examination Date:	2026-09-06 15:24:39 UTC	Forensic Engine:	Pushpak Indigenous Drone Forensics Toolkit (v1.0.0-phase1 (Build 2026.09))

EVIDENTIARY INTEGRITY 100% AUTHENTIC Dual SHA-256/BLAKE3 matched. Zero tampering.	AIRSPACE COMPLIANCE COMPLIANT (CLEAR) Operated in permitted boundaries.	FLIGHT ENVELOPE 0.1m 100m ALT Peak 45.0 km/h across 10 pts.	TERMINATION REASON RETURN-TO-HOME (RTH) Recorded flight disarm.
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Incident Narrative (Plain Language): On 2026-09-06, digital flight evidence for case **CASE-GUI-TEST-001** was acquired for forensic analysis. Physical and telemetry artifacts confirm the aircraft is a **ARDUPILOT** UAV (Model: *N/A*). The recorded flight operated between **2025-01-20 03:46:22 UTC** and **2025-01-20 03:46:31 UTC**, remaining airborne for a duration of **0.1 minutes** across **10 recorded positions**. The aircraft attained a peak altitude of **100.0 meters (328 ft AGL)** and a maximum ground speed of **45.0 km/h**. The aircraft operated within authorized flight bounds with no restricted boundary infringements detected. Flight conclusion was recorded as a **commanded Return-to-Home (RTH) procedure**.

Key Question for Legal / Judicial Review	Plain-Language Finding & Significance
1. Was the evidence altered or tampered with?	NO. Bitstream cryptographic verification (SHA-256 & BLAKE3) confirms 100% data authenticity with zero post-seizure tampering.
2. Did the drone enter restricted airspace?	NO. The flight remained fully within authorized, unrestricted airspace corridors.
3. Who or what was piloting the drone?	The flight operated primarily under AUTO, RTL , establishing active operator command and telemetry control.
4. What caused the flight to conclude?	Flight termination was triggered by Return-to-Home (RTH) .

PART II: TECHNICAL TELEMETRY & CRYPTOGRAPHIC EVIDENCE DOSSIER (EXPERT AUDIT)

2. EVIDENCE INTEGRITY & CHAIN OF CUSTODY

Source Evidence File:	C:\Users\pawan\AppData\Local\Temp\pytest-of-pawan\pytest-67\test_gui_api_analyze_evidence0\gui_test_flight.BIN
File Size:	1,715 bytes
SHA-256 Digest:	330f9a7eea6e16cb694c344913161372174e7624bf9066a855d46632049ef885
BLAKE3 Digest:	5ca5a1e38669da497980d724fcebca142b385459525a1a4646e9fd1609ebc6c9
Chain Verification:	VERIFIED INTACT (100% Cryptographic Match)

■ WHAT THIS MEANS IN PLAIN ENGLISH

Every electronic file has an unforgeable digital fingerprint called a cryptographic hash. Because both the SHA-256 and BLAKE3 mathematical digests match the baseline recorded at the time of seizure, the court and investigating officers can be 100% certain that this evidence has not been modified, edited, or fabricated by anyone since seizure.

3. FLIGHT MISSION ENVELOPE & TELEMETRY PROFILE

Platform / Architecture:	ARDUPILOT	Aircraft Model:	N/A
Hardware Serial No:	N/A	Total Telemetry Records:	37 events
Takeoff Time (UTC):	2025-01-20 03:46:22 UTC	Termination Time (UTC):	2025-01-20 03:46:31 UTC
Total Flight Duration:	9.0 seconds (0.1 mins)	Total GPS Waypoints:	10 points
Max Altitude (AGL):	100.0 meters (328.1 ft)	Peak Ground Velocity:	12.5 m/s (45.0 km/h)

■ WHAT THIS MEANS IN PLAIN ENGLISH

This section proves the physical flight capabilities and operating boundaries. The drone remained airborne for 0.1 minutes, reached a peak ground speed of 45.0 km/h, and climbed to 100.0 meters (328 ft AGL). These metrics legally establish the aircraft's operational range, airspeed, and chronological timeline.

4. FORENSIC ANOMALY FINDINGS & THREAT ASSESSMENT

Anomaly Type	Severity	Timestamp (UTC)	Location / Alt	Technical Finding Narrative
GPS SPOOFING OR TELEPORTATION	HIGH	03:46:23 UTC	19.0765, 72.8782 Alt: 20.0m	Impossible kinematic displacement: calculated velocity 76.5 m/s exceeds airframe limit (45.0 m/s) over 1.00s
GPS SPOOFING OR TELEPORTATION	HIGH	03:46:24 UTC	19.0770, 72.8787 Alt: 30.0m	Impossible kinematic displacement: calculated velocity 76.5 m/s exceeds airframe limit (45.0 m/s) over 1.00s
GPS SPOOFING OR TELEPORTATION	HIGH	03:46:25 UTC	19.0775, 72.8792 Alt: 40.0m	Impossible kinematic displacement: calculated velocity 76.5 m/s exceeds airframe limit (45.0 m/s) over 1.00s
GPS SPOOFING OR TELEPORTATION	HIGH	03:46:26 UTC	19.0780, 72.8797 Alt: 50.0m	Impossible kinematic displacement: calculated velocity 76.5 m/s exceeds airframe limit (45.0 m/s) over 1.00s
GPS SPOOFING OR TELEPORTATION	HIGH	03:46:27 UTC	19.0785, 72.8802 Alt: 60.0m	Impossible kinematic displacement: calculated velocity 76.5 m/s exceeds airframe limit (45.0 m/s) over 1.00s
GPS SPOOFING OR TELEPORTATION	HIGH	03:46:28 UTC	19.0790, 72.8807 Alt: 70.0m	Impossible kinematic displacement: calculated velocity 76.5 m/s exceeds airframe limit (45.0 m/s) over 1.00s
GPS SPOOFING OR TELEPORTATION	HIGH	03:46:29 UTC	19.0795, 72.8812 Alt: 80.0m	Impossible kinematic displacement: calculated velocity 76.5 m/s exceeds airframe limit (45.0 m/s) over 1.00s
GPS SPOOFING OR TELEPORTATION	HIGH	03:46:30 UTC	19.0800, 72.8817 Alt: 90.0m	Impossible kinematic displacement: calculated velocity 76.5 m/s exceeds airframe limit (45.0 m/s) over 1.00s
GPS SPOOFING OR TELEPORTATION	HIGH	03:46:31 UTC	19.0805, 72.8822 Alt: 100.0m	Impossible kinematic displacement: calculated velocity 76.5 m/s exceeds airframe limit (45.0 m/s) over 1.00s

■ WHAT THIS MEANS IN PLAIN ENGLISH

The automated forensic correlation engine identified 9 flight abnormalities. These findings establish whether the operator deliberately breached safety perimeters, disregarded critical low-power warnings, or experienced external interference during the flight.

5. FORENSIC CONFIDENCE & UNCERTAINTY ASSESSMENT

Overall Forensic Confidence Rating: HIGH (HIGH COURT ADMISSIBLE) - 99.0%
In accordance with ISO/IEC 27037:2012 guidelines, this assessment explicitly defines the mathematical certainty and evidentiary boundaries of the reconstructed flight profile.

Forensic Dimension	Confidence Level	Evidentiary Basis & Limitations
Bit-Stream Integrity	ABSOLUTE (100%)	Dual SHA-256 + BLAKE3 hashes match original media bit-stream.
Spatial Trajectory	HIGH (99.0%)	Avg Satellites: 14.0 Avg HDOP: 0.80 Multipath risk low.
Temporal Synchronization	HIGH (99.0%)	Correlated with GPS atomic UTC reference clock.
Evidence Limitations	DOCUMENTED	Barometric altitude uncorrected for local QNH pressure.

■ WHAT THIS MEANS IN PLAIN ENGLISH

For digital evidence to be legally admissible in court, its degree of precision must be established. Because the drone maintained continuous tracking with an average of 14.0 satellites and an HDOP rating of 0.80, the flight coordinates in this report are mathematically reliable to within approximately 1 to 2 meters of true physical position.

6. CHRONOLOGICAL MILESTONE TIMELINE

Timestamp (UTC)	Event Type	Mode / State	Forensic Evidence Context
2025-01-20 03:46:21	MODE CHANGE	AUTO	MODE
2025-01-20 03:46:21	ARM DISARM	AUTO	MSG
2025-01-20 03:46:22	GPS FIX	AUTO	GPS
2025-01-20 03:46:31	GPS FIX	AUTO	GPS
2025-01-20 03:46:32	RTH TRIGGER	AUTO	MSG
2025-01-20 03:46:32	MODE CHANGE	RTL	MODE
2025-01-20 03:46:36	ARM DISARM	RTL	MSG

■ WHAT THIS MEANS IN PLAIN ENGLISH

This timeline tracks the exact sequence of major decisions and events during the flight, such as motor startup, pilot control mode changes, automated alarms, and final touchdown, synchronized to international atomic time (UTC).

7. CERTIFICATE OF AUTHENTICITY (LEGAL ADMISSIBILITY)

I hereby certify under Section 63 of the Bharatiya Sakshya Adhiniyam, 2023 (corresponding to Section 65B of the Indian Evidence Act, 1872) and ISO/IEC 27037 standards that the electronic records, telemetry chronologies, and mathematical calculations contained within this report were generated by lawful digital forensic acquisition. The computing systems and software tools used to parse this evidence operated properly throughout the examination with zero alteration to original electronic memory. The cryptographic digests (SHA-256 and BLAKE3) confirm absolute bit-stream integrity of the exhibited evidence.

Examiner Signature: _____

Date: 2026-09-06

Name: Test Examiner

Title: Lead Digital Forensics Examiner

Agency: Cyber Forensic Investigation Laboratory (CFSL / State Police)

Official Seal: [AFFIX LABORATORY SEAL]